

OBJECTIF DE CE NOTEBOOK

Bienvenue dans l'outil plébiscité par les analystes de données Jupyter.

Il s'agit d'un outil permettant de mixer et d'alterner codes, textes et graphique.

Cet outil est formidable pour plusieurs raisons:

- il permet de tester des lignes de codes au fur et à mesure de votre rédaction, de constater immédiatement le résultat d'une instruction, de la corriger si nécessaire.
- De rédiger du texte pour expliquer l'approche suivie ou les résultats d'une analyse et de le mettre en forme grâce à du code html ou plus simple avec **Markdown**
- d'agrémenter de graphiques

Pour vous aider dans vos premiers pas à l'usage de Jupyter et de Python, nous avons rédigé ce notebook en vous indiquant les instructions à suivre.

Il vous suffit pour cela de saisir le code Python répondant à l'instruction donnée.

Vous verrez de temps à autre le code Python répondant à une instruction donnée mais cela est fait pour vous aider à comprendre la nature du travail qui vous est demandée.

Et garder à l'esprit, qu'il n'y a pas de solution unique pour résoudre un problème et qu'il y a autant de résolutions de problèmes que de développeurs ;)...

Note jeremy Est ce qu'il faut faire le calcul de la sous nutrition sur les pays qu'on a ? Est ce qu'il faut faire des graphiques ? Rajouter le soja La liste des céréales est difficile a trouver ...

```
#Importation de la librairie Pandas
import pandas as pd
import matplotlib.pyplot as plt

#Importation du fichier population.csv
population = pd.read_csv('./DAN-P4-FA0/population.csv')

#Importation du fichier dispo_alimentaire.csv
dispo_alimentaire = pd.read_csv('./DAN-P4-FA0/dispo_alimentaire.csv')

#Importation du fichier aide_alimentaire.csv
aide_alimentaire = pd.read_csv('./DAN-P4-FA0/aide_alimentaire.csv')

#Importation du fichier sous_nutrition.csv
sous_nutrition = pd.read_csv('./DAN-P4-FA0/sous_nutrition.csv')

#Afficher les dimensions du dataset
print("Le tableau comporte {} observation(s) ou  
article(s)".format(population.shape[0]))
print("Le tableau comporte {} colonne(s)".format(population.shape[1]))
```

Le tableau comporte 1416 observation(s) ou article(s)
Le tableau comporte 3 colonne(s)

```
print(population.columns)
print("-----")
print(population.dtypes)
print("-----")
print(population.info())
```

```
Index(['Zone', 'Année', 'Valeur'], dtype='object')
```

```
-----
Zone      object
Année     int64
Valeur    float64
dtype: object
-----
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1416 entries, 0 to 1415
Data columns (total 3 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   Zone        1416 non-null   object
1   Année       1416 non-null   int64
2   Valeur      1416 non-null   float64
dtypes: float64(1), int64(1), object(1)
memory usage: 33.3+ KB
None
```

```
population.head()
```

```
{"columns":[{"name":"index","rawType":"int64","type":"integer"},
{"name":"Zone","rawType":"object","type":"string"},
{"name":"Année","rawType":"int64","type":"integer"},
{"name":"Valeur","rawType":"float64","type":"float"}],"conversionMethod":"pd.DataFrame","ref":"8b65df99-6ef7-4527-9aa5-d860f1769240","rows":
[["0","Afghanistan","2013","32269.589"],
["1","Afghanistan","2014","33370.794"],
["2","Afghanistan","2015","34413.603"],
["3","Afghanistan","2016","35383.032"],
["4","Afghanistan","2017","36296.113"]],"shape":
{"columns":3,"rows":5}}
```

#Nous allons harmoniser les unités. Pour cela, nous avons décidé de multiplier la population par 1000

#Multiplication de la colonne valeur par 1000

```
population['Valeur']*= 1000
```

#changement du nom de la colonne Valeur par Population

```
population.rename(columns={'Valeur':'Population'}, inplace=True)
```

```
#Affichage les 5 premières lignes de la table pour voir les modifications
```

```
population.head()
```

```
{"columns":[{"name":"index","rawType":"int64","type":"integer"},
{"name":"Zone","rawType":"object","type":"string"},
{"name":"Année","rawType":"int64","type":"integer"},
{"name":"Population","rawType":"float64","type":"float"}],"conversionMethod":"pd.DataFrame","ref":"8b6edff8-6df6-46cf-855a-b2bd33b4acd0","rows":[["0","Afghanistan","2013","32269589.0"],
["1","Afghanistan","2014","33370794.0"],
["2","Afghanistan","2015","34413603.0"],
["3","Afghanistan","2016","35383032.0"],
["4","Afghanistan","2017","36296113.0"]],"shape":
{"columns":3,"rows":5}}
```

```
#Afficher les dimensions du dataset
```

```
print("Le tableau comporte {} observation(s) ou article(s)".format(dispo_alimentaire.shape[0]))
```

```
print("Le tableau comporte {} colonne(s)".format(dispo_alimentaire.shape[1]))
```

```
Le tableau comporte 15605 observation(s) ou article(s)
```

```
Le tableau comporte 18 colonne(s)
```

```
#Consulter le nombre de colonnes
```

```
dispo_alimentaire.columns
```

```
Index(['Zone', 'Produit', 'Origine', 'Aliments pour animaux',
       'Autres Utilisations', 'Disponibilité alimentaire
(Kcal/personne/jour)',
       'Disponibilité alimentaire en quantité (kg/personne/an)',
       'Disponibilité de matière grasse en quantité
(g/personne/jour)',
       'Disponibilité de protéines en quantité (g/personne/jour)',
       'Disponibilité intérieure', 'Exportations - Quantité',
       'Importations - Quantité', 'Nourriture', 'Pertes',
       'Production',
       'Semences', 'Traitement', 'Variation de stock'],
      dtype='object')
```

```
#Affichage les 5 premières lignes de la table
```

```
dispo_alimentaire.head()
```

```
{"columns":[{"name":"index","rawType":"int64","type":"integer"},
{"name":"Zone","rawType":"object","type":"string"},
{"name":"Produit","rawType":"object","type":"string"},
{"name":"Origine","rawType":"object","type":"string"},
{"name":"Aliments pour animaux","rawType":"float64","type":"float"},
{"name":"Autres Utilisations","rawType":"float64","type":"float"},
{"name":"Disponibilité alimentaire
```

```
(Kcal/personne/jour)", "rawType": "float64", "type": "float"},
{"name": "Disponibilité alimentaire en quantité
(kg/personne/an)", "rawType": "float64", "type": "float"},
{"name": "Disponibilité de matière grasse en quantité
(g/personne/jour)", "rawType": "float64", "type": "float"},
{"name": "Disponibilité de protéines en quantité
(g/personne/jour)", "rawType": "float64", "type": "float"},
{"name": "Disponibilité
intérieure", "rawType": "float64", "type": "float"}, {"name": "Exportations
- Quantité", "rawType": "float64", "type": "float"}, {"name": "Importations
- Quantité", "rawType": "float64", "type": "float"},
{"name": "Nourriture", "rawType": "float64", "type": "float"},
{"name": "Pertes", "rawType": "float64", "type": "float"},
{"name": "Production", "rawType": "float64", "type": "float"},
{"name": "Semences", "rawType": "float64", "type": "float"},
{"name": "Traitement", "rawType": "float64", "type": "float"},
{"name": "Variation de
stock", "rawType": "float64", "type": "float"}], "conversionMethod": "pd.Dat
aFrame", "ref": "3a6a49f7-3c3a-45f5-a2d3-325692aa3fd6", "rows":
[["0", "Afghanistan", "Abats
Comestible", "animale", null, null, "5.0", "1.72", "0.2", "0.77", "53.0", null,
null, "53.0", null, "53.0", null, null, null], ["1", "Afghanistan", "Agrumes,
Autres", "vegetale", null, null, "1.0", "1.29", "0.01", "0.02", "41.0", "2.0", "
40.0", "39.0", "2.0", "3.0", null, null, null], ["2", "Afghanistan", "Aliments
pour
enfants", "vegetale", null, null, "1.0", "0.06", "0.01", "0.03", "2.0", null, "2
.0", "2.0", null, null, null, null, null],
["3", "Afghanistan", "Ananas", "vegetale", null, null, "0.0", "0.0", null, null
, "0.0", null, "0.0", "0.0", null, null, null, null],
["4", "Afghanistan", "Bananes", "vegetale", null, null, "4.0", "2.7", "0.02", "
0.05", "82.0", null, "82.0", "82.0", null, null, null, null, null]], "shape":
{"columns": 18, "rows": 5}}
```

```
#remplacement des NaN dans le dataset par des 0
dispo_alimentaire.fillna(0, inplace=True)
```

```
dispo_alimentaire.describe().T
```

```
{"columns": [{"name": "index", "rawType": "object", "type": "string"},
{"name": "count", "rawType": "float64", "type": "float"},
{"name": "mean", "rawType": "float64", "type": "float"},
{"name": "std", "rawType": "float64", "type": "float"},
{"name": "min", "rawType": "float64", "type": "float"},
{"name": "25%", "rawType": "float64", "type": "float"},
{"name": "50%", "rawType": "float64", "type": "float"},
{"name": "75%", "rawType": "float64", "type": "float"},
{"name": "max", "rawType": "float64", "type": "float"}], "conversionMethod":
"pd.DataFrame", "ref": "1838df39-032f-48f4-96b9-2929c3e713fb", "rows":
[["Aliments pour
animaux", "15605.0", "83.57866068567766", "1779.2891527729462", "0.0", "0.0
```

```

", "0.0", "0.0", "150000.0"], ["Autres
Utilisations", "15605.0", "55.43242550464595", "3013.6298116440717", "0.0",
"0.0", "0.0", "0.0", "347309.0"], ["Disponibilité alimentaire
(Kcal/personne/jour)", "15605.0", "31.748926626081385", "102.961193917252
12", "-21.0", "0.0", "3.0", "18.0", "1711.0"], ["Disponibilité alimentaire
en quantité
(kg/personne/an)", "15605.0", "7.8309477731496315", "23.478909452634422",
"-1.93", "0.01", "0.52", "4.27", "430.76"], ["Disponibilité de matière
grasse en quantité
(g/personne/jour)", "15605.0", "0.9697539250240308", "3.2466949579777937",
"-0.03", "0.0", "0.02", "0.31", "60.76"], ["Disponibilité de protéines en
quantité
(g/personne/jour)", "15605.0", "0.9065126561999358", "3.143515762577603",
"-0.37", "0.0", "0.02", "0.35", "54.97"], ["Disponibilité
intérieure", "15605.0", "631.1434796539571", "9002.563665606789", "-
3430.0", "0.0", "6.0", "73.0", "739267.0"], ["Exportations -
Quantité", "15605.0", "86.64902274911887", "933.435207725959", "-
41.0", "0.0", "0.0", "4.0", "42797.0"], ["Importations -
Quantité", "15605.0", "83.05370073694328", "700.0994508124281", "-
201.0", "0.0", "1.0", "16.0", "63381.0"],
["Nourriture", "15605.0", "312.4804870233899", "4242.8547243817975", "-
246.0", "0.0", "3.0", "40.0", "426850.0"],
["Pertes", "15605.0", "29.073886574815763", "584.6722357778441", "0.0", "0.
0", "0.0", "0.0", "55047.0"],
["Production", "15605.0", "641.44056392182", "9270.864198432677", "0.0", "0
.0", "0.0", "42.0", "739267.0"],
["Semences", "15605.0", "9.912271707785965", "194.89784342889894", "0.0", "
0.0", "0.0", "0.0", "17060.0"],
["Traitement", "15605.0", "141.2808074335149", "3992.5676214000828", "-
19.0", "0.0", "0.0", "0.0", "326711.0"], ["Variation de stock", "15605.0", "-
6.690291573213713", "362.3807871916048", "-
39863.0", "0.0", "0.0", "0.0", "5284.0"]], "shape":{"columns":8,"rows":15}}

```

```

#multiplication de toutes les lignes contenant des milliers de tonnes
en Kg

```

```

COLONNESDISPO = ['Aliments pour animaux', 'Autres
Utilisations', 'Disponibilité intérieure', 'Exportations -
Quantité', 'Importations - Quantité',

```

```

'Nourriture', 'Pertes', 'Production', 'Semences', 'Traitement', 'Variation
de stock']

```

```

for i in COLONNESDISPO:
    dispo_alimentaire[i]*=1000

```

```

#Affichage les 5 premières lignes de la table
dispo_alimentaire.head()

```

```

{"columns":[{"name":"index","rawType":"int64","type":"integer"},
{"name":"Zone","rawType":"object","type":"string"},

```

```
{
  "name": "Produit", "rawType": "object", "type": "string",
  "name": "Origine", "rawType": "object", "type": "string",
  "name": "Aliments pour animaux", "rawType": "float64", "type": "float",
  "name": "Autres Utilisations", "rawType": "float64", "type": "float",
  "name": "Disponibilité alimentaire
(Kcal/personne/jour)", "rawType": "float64", "type": "float",
  "name": "Disponibilité alimentaire en quantité
(kg/personne/an)", "rawType": "float64", "type": "float",
  "name": "Disponibilité de matière grasse en quantité
(g/personne/jour)", "rawType": "float64", "type": "float",
  "name": "Disponibilité de protéines en quantité
(g/personne/jour)", "rawType": "float64", "type": "float",
  "name": "Disponibilité
intérieure", "rawType": "float64", "type": "float"},
{"name": "Exportations
- Quantité", "rawType": "float64", "type": "float"},
{"name": "Importations
- Quantité", "rawType": "float64", "type": "float"},
{"name": "Nourriture", "rawType": "float64", "type": "float"},
{"name": "Pertes", "rawType": "float64", "type": "float"},
{"name": "Production", "rawType": "float64", "type": "float"},
{"name": "Semences", "rawType": "float64", "type": "float"},
{"name": "Traitement", "rawType": "float64", "type": "float"},
{"name": "Variation de
stock", "rawType": "float64", "type": "float"}],
"conversionMethod": "pd.DataFrame",
"ref": "fb3bad59-5940-42ca-b699-79a92435fcac",
"rows": [
  ["0", "Afghanistan", "Abats
Comestible", "animale", "0.0", "0.0", "5.0", "1.72", "0.2", "0.77", "53000.0",
"0.0", "0.0", "53000.0", "0.0", "53000.0", "0.0", "0.0", "0.0"],
  ["1", "Afghanistan", "Agrumes,
Autres", "vegetale", "0.0", "0.0", "1.0", "1.29", "0.01", "0.02", "41000.0", "2
000.0", "40000.0", "39000.0", "2000.0", "3000.0", "0.0", "0.0", "0.0"],
  ["2", "Afghanistan", "Aliments pour
enfants", "vegetale", "0.0", "0.0", "1.0", "0.06", "0.01", "0.03", "2000.0", "0
.0", "2000.0", "2000.0", "0.0", "0.0", "0.0", "0.0", "0.0"],
  ["3", "Afghanistan", "Ananas", "vegetale", "0.0", "0.0", "0.0", "0.0", "0.0", "
0.0", "0.0", "0.0", "0.0", "0.0", "0.0", "0.0", "0.0", "0.0"],
  ["4", "Afghanistan", "Bananes", "vegetale", "0.0", "0.0", "4.0", "2.7", "0.02
", "0.05", "82000.0", "0.0", "82000.0", "82000.0", "0.0", "0.0", "0.0", "0.0", "0
.0"]],
"shape": {"columns": 18, "rows": 5}}

```

```
#Afficher les dimensions du dataset
print("Le tableau comporte {} observation(s) ou
article(s)".format(aide_alimentaire.shape[0]))
print("Le tableau comporte {}
colonne(s)".format(aide_alimentaire.shape[1]))

```

```
Le tableau comporte 1475 observation(s) ou article(s)
Le tableau comporte 4 colonne(s)

```

```
#Consulter le nombre de colonnes
aide_alimentaire.columns

```

```
Index(['Pays bénéficiaire', 'Année', 'Produit', 'Valeur'],
      dtype='object')
```

```
#Affichage les 5 premières lignes de la table
aide_alimentaire.head()
```

```
{ "columns": [{"name": "index", "rawType": "int64", "type": "integer"},
{"name": "Pays bénéficiaire", "rawType": "object", "type": "string"},
{"name": "Année", "rawType": "int64", "type": "integer"},
{"name": "Produit", "rawType": "object", "type": "string"},
{"name": "Valeur", "rawType": "int64", "type": "integer"}], "conversionMethod": "pd.DataFrame", "ref": "a27d97d4-165a-43c9-92ea-99dbc4befb24", "rows":
[[ "0", "Afghanistan", "2013", "Autres non-céréales", "682"],
[ "1", "Afghanistan", "2014", "Autres non-céréales", "335"],
[ "2", "Afghanistan", "2013", "Blé et Farin", "39224"],
[ "3", "Afghanistan", "2014", "Blé et Farin", "15160"],
[ "4", "Afghanistan", "2013", "Céréales", "40504"]], "shape":
{ "columns": 4, "rows": 5}}
```

```
#changement du nom de la colonne Pays bénéficiaire par Zone
aide_alimentaire.rename(columns={'Pays bénéficiaire': 'Zone'},
inplace=True)
aide_alimentaire.rename(columns={'Valeur': 'Aide Alimentaire'},
inplace=True)
```

```
#Multiplication de la colonne Aide_alimentaire qui contient des tonnes
par 1000 pour avoir des kg
aide_alimentaire['Aide Alimentaire'] *= 1000
```

```
#Affichage les 5 premières lignes de la table
aide_alimentaire.head()
```

```
{ "columns": [{"name": "index", "rawType": "int64", "type": "integer"},
{"name": "Zone", "rawType": "object", "type": "string"},
{"name": "Année", "rawType": "int64", "type": "integer"},
{"name": "Produit", "rawType": "object", "type": "string"}, {"name": "Aide
Alimentaire", "rawType": "int64", "type": "integer"}], "conversionMethod": "
pd.DataFrame", "ref": "f3a9f347-3d16-4f93-b0ab-3556b7c4a347", "rows":
[[ "0", "Afghanistan", "2013", "Autres non-céréales", "682000"],
[ "1", "Afghanistan", "2014", "Autres non-céréales", "335000"],
[ "2", "Afghanistan", "2013", "Blé et Farin", "39224000"],
[ "3", "Afghanistan", "2014", "Blé et Farin", "15160000"],
[ "4", "Afghanistan", "2013", "Céréales", "40504000"]], "shape":
{ "columns": 4, "rows": 5}}
```

```
#Afficher les dimensions du dataset
print("Le tableau comporte {} observation(s) ou
article(s)".format(sous_nutrition.shape[0]))
print("Le tableau comporte {}
colonne(s)".format(sous_nutrition.shape[1]))
```

Le tableau comporte 1218 observation(s) ou article(s)
Le tableau comporte 3 colonne(s)

#Consulter le nombre de colonnes

```
sous_nutrition.columns
```

```
Index(['Zone', 'Année', 'Valeur'], dtype='object')
```

#Afficher les 5 premières lignes de la table

```
sous_nutrition.head()
```

```
{"columns":[{"name":"index","rawType":"int64","type":"integer"},
{"name":"Zone","rawType":"object","type":"string"},
{"name":"Année","rawType":"object","type":"string"},
{"name":"Valeur","rawType":"object","type":"string"}],"conversionMethod":
"pd.DataFrame","ref":"1922078c-6a0b-411d-9c28-b57ddac69ce9","rows":
[[{"0","Afghanistan","2012-2014","8.6"},{"1","Afghanistan","2013-
2015","8.8"},{"2","Afghanistan","2014-2016","8.9"},
{"3","Afghanistan","2015-2017","9.7"},{"4","Afghanistan","2016-
2018","10.5"}],"shape":{"columns":3,"rows":5}}
```

```
sous_nutrition.describe().T
```

```
{"columns":[{"name":"index","rawType":"object","type":"string"},
{"name":"count","rawType":"object","type":"unknown"},
{"name":"unique","rawType":"object","type":"unknown"},
{"name":"top","rawType":"object","type":"string"},
{"name":"freq","rawType":"object","type":"unknown"}],"conversionMethod":
"pd.DataFrame","ref":"4e8fa91a-c93d-406a-9bc0-b60676a51540","rows":
[["Zone","1218","203","Afghanistan","6"],["Année","1218","6","2012-
2014","203"],["Valeur","624","139","<0.1","120"]],"shape":
{"columns":4,"rows":3}}
```

#Conversion de la colonne sous nutrition en numérique

```
sous_nutrition['Valeur'] = pd.to_numeric(sous_nutrition['Valeur'],
errors='coerce')
```

#Conversion de la colonne (avec l'argument errors=coerce qui permet de convertir automatiquement les lignes qui ne sont pas des nombres en NaN)

#Puis remplacement des NaN en 0

```
sous_nutrition.fillna(0, inplace=True)
```

#changement du nom de la colonne Valeur par sous_nutrition

```
sous_nutrition.rename(columns={'Valeur':'sous_nutrition'},
inplace=True)
```

#Multiplication de la colonne sous_nutrition par 1000000

```
sous_nutrition['sous_nutrition'] *= 1000000
```

```

#Afficher les 5 premières lignes de la table
sous_nutrition.head()

{"columns":[{"name":"index","rawType":"int64","type":"integer"},
{"name":"Zone","rawType":"object","type":"string"},
{"name":"Année","rawType":"object","type":"string"},
{"name":"sous_nutrition","rawType":"float64","type":"float"}],"conversionMethod":"pd.DataFrame","ref":"9eb57882-c7fc-4406-8288-26c8eceb9e39","rows":[["0","Afghanistan","2012-2014","8600000.0"],
["1","Afghanistan","2013-2015","8800000.0"],["2","Afghanistan","2014-2016","8900000.0"],["3","Afghanistan","2015-2017","9700000.0"],
["4","Afghanistan","2016-2018","10500000.0"]],"shape":
{"columns":3,"rows":5}}

# Il faut tout d'abord faire une jointure entre la table population et
la table sous nutrition, en ciblant l'année 2017
population_2017 = population.loc[population['Année']==2017, ['Zone',
'Population']]

#Affichage du dataset
population_2017.head()

{"columns":[{"name":"index","rawType":"int64","type":"integer"},
{"name":"Zone","rawType":"object","type":"string"},
{"name":"Population","rawType":"float64","type":"float"}],"conversionMethod":"pd.DataFrame","ref":"7519a54f-4401-45f6-9163-b850bd11220e","rows":[["4","Afghanistan","36296113.0"],["10","Afrique du Sud","57009756.0"],["16","Albanie","2884169.000000001"],
["22","Algérie","41389189.0"],
["28","Allemagne","82658409.0"]],"shape":{"columns":2,"rows":5}}

#Calcul et affichage du nombre de personnes en état de sous nutrition

jointure = pd.merge(
    population.loc[population['Année'] == 2017, ['Zone',
'Population']],
    sous_nutrition.loc[sous_nutrition['Année'] == '2016-2018',
['Zone', 'sous_nutrition']],
    on='Zone'
)

personnes_en_sous_nutrition = round(jointure['sous_nutrition'].sum())
proportions_sous_nutrition = round(jointure['sous_nutrition'].sum() *
100 / jointure['Population'].sum(), 2)

print(f'Il y a {personnes_en_sous_nutrition} personnes en état de sous
nutrition en 2017')
print(f'Proportion de personnes en état de sous nutrition :
{proportions_sous_nutrition} %')

```

Il y a 535700000 personnes en état de sous nutrition en 2017
Proportion de personnes en état de sous nutrition : 7.1 %

#Combien mange en moyenne un être humain ? Source => 2500Kcal, source FAO/STAT

calories_humain = 2500

#On commence par faire une jointure entre le data frame population et Dispo_alimentaire afin d'ajouter dans ce dernier la population
population_alimentaire = pd.merge(dispo_alimentaire, population_2017, on = 'Zone')

#Affichage du nouveau dataframe

population_alimentaire.head()

```
{"columns":[{"name":"index","rawType":"int64","type":"integer"},
{"name":"Zone","rawType":"object","type":"string"},
{"name":"Produit","rawType":"object","type":"string"},
{"name":"Origine","rawType":"object","type":"string"},
{"name":"Aliments pour animaux","rawType":"float64","type":"float"},
{"name":"Autres Utilisations","rawType":"float64","type":"float"},
{"name":"Disponibilité alimentaire
(Kcal/personne/jour)","rawType":"float64","type":"float"},
{"name":"Disponibilité alimentaire en quantité
(kg/personne/an)","rawType":"float64","type":"float"},
{"name":"Disponibilité de matière grasse en quantité
(g/personne/jour)","rawType":"float64","type":"float"},
{"name":"Disponibilité de protéines en quantité
(g/personne/jour)","rawType":"float64","type":"float"},
{"name":"Disponibilité
intérieure","rawType":"float64","type":"float"},{"name":"Exportations
- Quantité","rawType":"float64","type":"float"},{"name":"Importations
- Quantité","rawType":"float64","type":"float"},
{"name":"Nourriture","rawType":"float64","type":"float"},
{"name":"Pertes","rawType":"float64","type":"float"},
{"name":"Production","rawType":"float64","type":"float"},
{"name":"Semences","rawType":"float64","type":"float"},
{"name":"Traitement","rawType":"float64","type":"float"},
{"name":"Variation de stock","rawType":"float64","type":"float"},
{"name":"Population","rawType":"float64","type":"float"}],"conversionM
ethod":"pd.DataFrame","ref":"a3bcac7d-0d9b-4c5d-a7d0-
ec1687bfff43","rows":[["0","Afghanistan","Abats
Comestible","animale","0.0","0.0","5.0","1.72","0.2","0.77","53000.0",
"0.0","0.0","53000.0","0.0","53000.0","0.0","0.0","0.0","36296113.0"],
["1","Afghanistan","Agrumes,
Autres","vegetale","0.0","0.0","1.0","1.29","0.01","0.02","41000.0","2
000.0","40000.0","39000.0","2000.0","3000.0","0.0","0.0","0.0","362961
13.0"],["2","Afghanistan","Aliments pour
enfants","vegetale","0.0","0.0","1.0","0.06","0.01","0.03","2000.0","0
.0","2000.0","2000.0","0.0","0.0","0.0","0.0","0.0","36296113.0"]]
```

```
[ "3", "Afghanistan", "Ananas", "vegetale", "0.0", "0.0", "0.0", "0.0", "0.0", "0.0", "0.0", "0.0", "0.0", "0.0", "0.0", "0.0", "0.0", "0.0", "36296113.0" ],
[ "4", "Afghanistan", "Bananes", "vegetale", "0.0", "0.0", "4.0", "2.7", "0.02", "0.05", "82000.0", "0.0", "82000.0", "82000.0", "0.0", "0.0", "0.0", "0.0", "0.0", "36296113.0" ] ], "shape": {"columns": 19, "rows": 5}}
```

#Création de la colonne dispo_kcal avec calcul des kcal disponibles mondialement

```
population_alimentaire['dispo_kcal'] =
population_alimentaire['Disponibilité alimentaire
(Kcal/personne/jour)'] * population_alimentaire['Population']
population_alimentaire['dispo_kcal']
```

```
0      181480565.0
1      36296113.0
2      36296113.0
3           0.0
4     145184452.0
```

```
...
15411    28621755.0
15412    6996429.0
15413         0.0
15414         0.0
15415    2544156.0
```

Name: dispo_kcal, Length: 15416, dtype: float64

#Calcul du nombre d'humains pouvant être nourris

```
nb_personnes_pouvant_etre_nourris =
round(population_alimentaire['dispo_kcal'].sum()/calories_humain)
prop_personnes_pouvant_etre_nourris =
round(nb_personnes_pouvant_etre_nourris*100/population_2017['Population'].sum(),2)
print('Total humains pouvant être nourris :',
nb_personnes_pouvant_etre_nourris)
print('Proportions humains pouvant être nourris :',
prop_personnes_pouvant_etre_nourris, '%')
```

Total humains pouvant être nourris : 8367593851

Proportions humains pouvant être nourris : 110.86 %

#Transfert des données avec les végétaux dans un nouveau dataframe

```
vegetaux =
population_alimentaire.loc[population_alimentaire['Origine']=='vegetale',:]
vegetaux.head()
```

```
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{"name": "Zone", "rawType": "object", "type": "string"},
{"name": "Produit", "rawType": "object", "type": "string"},
{"name": "Origine", "rawType": "object", "type": "string"},
```

```

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{"name": "Autres Utilisations", "rawType": "float64", "type": "float"},
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{"name": "Disponibilité alimentaire en quantité
(kg/personne/an)", "rawType": "float64", "type": "float"},
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(g/personne/jour)", "rawType": "float64", "type": "float"},
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(g/personne/jour)", "rawType": "float64", "type": "float"},
{"name": "Disponibilité
intérieure", "rawType": "float64", "type": "float"}, {"name": "Exportations
- Quantité", "rawType": "float64", "type": "float"}, {"name": "Importations
- Quantité", "rawType": "float64", "type": "float"},
{"name": "Nourriture", "rawType": "float64", "type": "float"},
{"name": "Pertes", "rawType": "float64", "type": "float"},
{"name": "Production", "rawType": "float64", "type": "float"},
{"name": "Semences", "rawType": "float64", "type": "float"},
{"name": "Traitement", "rawType": "float64", "type": "float"},
{"name": "Variation de stock", "rawType": "float64", "type": "float"},
{"name": "Population", "rawType": "float64", "type": "float"},
{"name": "dispo_kcal", "rawType": "float64", "type": "float"}], "conversionM
ethod": "pd.DataFrame", "ref": "6bc5ac19-3816-4014-9f48-
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Autres", "vegetale", "0.0", "0.0", "1.0", "1.29", "0.01", "0.02", "41000.0", "2
000.0", "40000.0", "39000.0", "2000.0", "3000.0", "0.0", "0.0", "0.0", "362961
13.0", "36296113.0"], [2, "Afghanistan", "Aliments pour
enfants", "vegetale", "0.0", "0.0", "1.0", "0.06", "0.01", "0.03", "2000.0", "0
.0", "2000.0", "2000.0", "0.0", "0.0", "0.0", "0.0", "0.0", "36296113.0", "3629
6113.0"],
[3, "Afghanistan", "Ananas", "vegetale", "0.0", "0.0", "0.0", "0.0", "0.0", "
0.0", "0.0", "0.0", "0.0", "0.0", "0.0", "0.0", "0.0", "36296113.0
", "0.0"],
[4, "Afghanistan", "Bananes", "vegetale", "0.0", "0.0", "4.0", "2.7", "0.02"
, "0.05", "82000.0", "0.0", "82000.0", "82000.0", "0.0", "0.0", "0.0", "0.0", "0
.0", "36296113.0", "145184452.0"],
[6, "Afghanistan", "Bière", "vegetale", "0.0", "0.0", "0.0", "0.09", "0.0", "
0.0", "3000.0", "0.0", "3000.0", "3000.0", "0.0", "0.0", "0.0", "0.0", "0.0", "3
6296113.0", "0.0"]], "shape": {"columns": 20, "rows": 5}}

```

#Calcul du nombre de kcal disponible pour les végétaux

```
kcal_vegetaux = round(vegetaux['dispo_kcal'].sum())
```

```
kcal_vegetaux
```

```
17260764211501
```

#Calcul du nombre d'humains pouvant être nourris avec les végétaux

```
nb_humains_nourris_par_vegetaux = round(kcal_vegetaux/calories_humain)
```

```
prop_humains_nourris_par_vegetaux =
```

```
round(nb_humains_nourris_par_vegetaux*100/population_2017['Population']
```

```

].sum(), 2)
print('Total humains pouvant être nourris avec les végétaux:',
nb_humains_nourris_par_vegetaux)
print('Proportions humains pouvant être nourris avec les végétaux :',
prop_humains_nourris_par_vegetaux, '%')

Total humains pouvant être nourris avec les végétaux: 6904305685
Proportions humains pouvant être nourris avec les végétaux : 91.47 %

#Calcul de la disponibilité totale
dispo_interieure_totale = round(population_alimentaire['Disponibilité
intérieure'].sum())
dispo_interieure_totale

9733927000

#création d'une boucle for pour afficher les différentes valeurs en
fonction des colonnes aliments pour animaux, pertes, nourritures,
Liste_utilisation_produits = ['Aliments pour animaux', 'Pertes',
'Nourriture', 'Semences', 'Traitement', 'Autres Utilisations']

for i in Liste_utilisation_produits:
    print('Proportion de', i, ':',
round(population_alimentaire[i].sum()*100/dispo_interieure_totale, 2),
'%')

Proportion de Aliments pour animaux : 13.23 %
Proportion de Pertes : 4.65 %
Proportion de Nourriture : 49.37 %
Proportion de Semences : 1.58 %
Proportion de Traitement : 22.45 %
Proportion de Autres Utilisations : 8.82 %

#Création d'une liste avec toutes les variables
liste_cereales= ["Blé", "Riz (Eq Blanchi)", "Orge", "Maïs", "Seigle",
"Avoine", "Millet", "Sorgho", "Céréales, Autres"]

#Création d'un dataframe avec les informations uniquement pour ces
céréales
cereales =
population_alimentaire.loc[population_alimentaire['Produit'].isin(list
e_cereales),:]
cereales.head()

{"columns":[{"name":"index","rawType":"int64","type":"integer"},
{"name":"Zone","rawType":"object","type":"string"},
{"name":"Produit","rawType":"object","type":"string"},
{"name":"Origine","rawType":"object","type":"string"},
{"name":"Aliments pour animaux","rawType":"float64","type":"float"},
{"name":"Autres Utilisations","rawType":"float64","type":"float"},
{"name":"Disponibilité alimentaire

```



```
#Affichage de la proportion d'alimentation humaine
prop_alimentation_humaine=
round(cereales['Nourriture'].sum()*100/dispo_interieure_cereales_total
e,2)
print("Proportion d'alimentation humaine :",
prop_alimentation_humaine, "%")
```

Proportion d'alimentation humaine : 42.91 %

```
#Création de la colonne proportion par pays
jointure['Prop_sous_alim_par_pays'] =
jointure['sous_nutrition']*100/jointure['Population']
jointure.head()
```

```
{"columns":[{"name":"index","rawType":"int64","type":"integer"},
{"name":"Zone","rawType":"object","type":"string"},
{"name":"Population","rawType":"float64","type":"float"},
{"name":"sous_nutrition","rawType":"float64","type":"float"},
{"name":"Prop_sous_alim_par_pays","rawType":"float64","type":"float"}]
,"conversionMethod":"pd.DataFrame","ref":"b49292ef-04fc-4610-a80d-
e94f4e3713e5","rows":
[["0","Afghanistan","36296113.0","10500000.0","28.928717518594897"],
["1","Afrique du Sud","57009756.0","3100000.0","5.437665791798864"],
["2","Albanie","2884169.0000000001","100000.0","3.4672032048052652"],
["3","Algérie","41389189.0","1300000.0","3.140916822506476"],
["4","Allemagne","82658409.0","0.0","0.0"]],"shape":
{"columns":4,"rows":5}}
```

```
#affichage après trie des 10 pires pays
jointure.sort_values(by = ['Prop_sous_alim_par_pays'],
ascending=False).head(10)
```

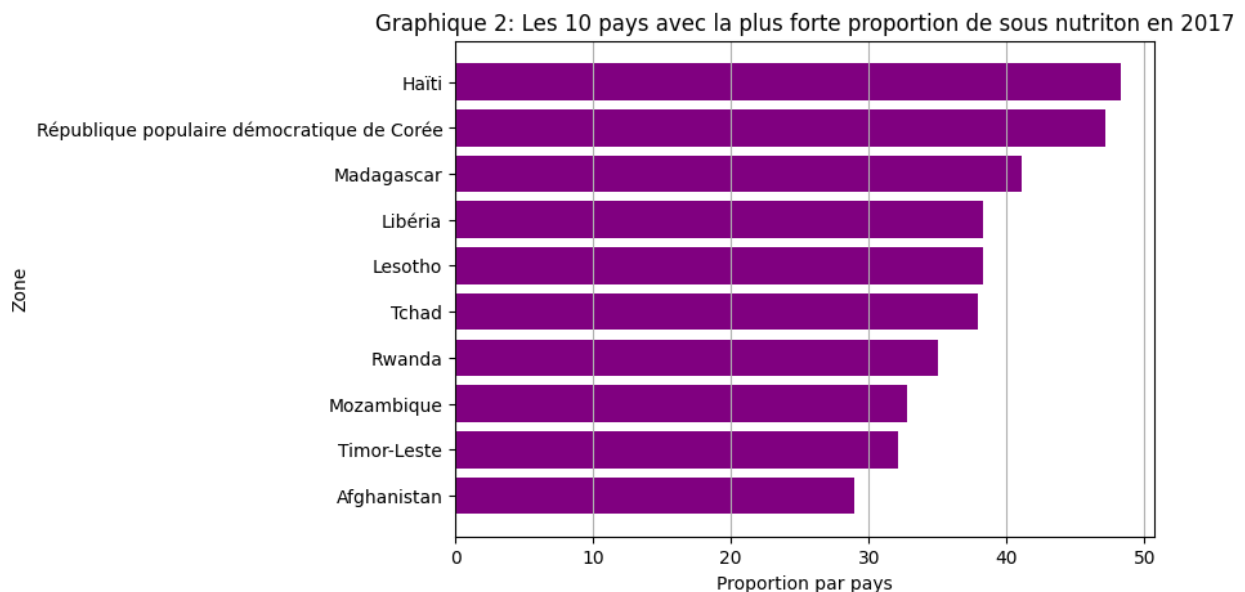
```
{"columns":[{"name":"index","rawType":"int64","type":"integer"},
{"name":"Zone","rawType":"object","type":"string"},
{"name":"Population","rawType":"float64","type":"float"},
{"name":"sous_nutrition","rawType":"float64","type":"float"},
{"name":"Prop_sous_alim_par_pays","rawType":"float64","type":"float"}]
,"conversionMethod":"pd.DataFrame","ref":"09612fa7-f9b1-48d7-97af-
fa5b7457fda9","rows":
[["78","Haïti","10982366.0000000002","5300000.0","48.25918203782317"],
["157","République populaire démocratique de
Corée","25429825.0","12000000.0","47.188684939829514"],
["108","Madagascar","25570512.0","10500000.0","41.06292435599256"],
["103","Libéria","4702226.0","1800000.0","38.27974240285346"],
["100","Lesotho","2091534.0","800000.0","38.24943797232079"],
["183","Tchad","15016753.0","5700000.0","37.95760641464903"],
["161","Rwanda","11980961.0","4200000.0","35.05561866030613"],
["121","Mozambique","28649018.0","9400000.0","32.81089774176553"],
["186","Timor-Leste","1243258.0","400000.0","32.17353115765191"],
```

```

["0","Afghanistan","36296113.0","10500000.0","28.928717518594897"]], "shape":{"columns":4,"rows":10}}

graphique= jointure.sort_values(by = ['Prop_sous_alim_par_pays'],
ascending= False).head(10)
plt.figure(figsize=(7,5))
plt.barh(graphique['Zone'], graphique['Prop_sous_alim_par_pays'],
color = 'purple')
plt.xticks(fontsize = 10)
plt.yticks(fontsize = 10)
plt.grid(axis='x')
plt.title('Graphique 2: Les 10 pays avec la plus forte proportion de
sous nutrion en 2017')
plt.ylabel('Zone', fontsize = 10)
plt.xlabel('Proportion par pays', fontsize = 10)
plt.gca().invert_yaxis()

```



```

#calcul du total de l'aide alimentaire par pays
total_aide_alimentaire = aide_alimentaire.groupby(['Zone'])['Aide
Alimentaire'].sum().reset_index()
total_aide_alimentaire.head()

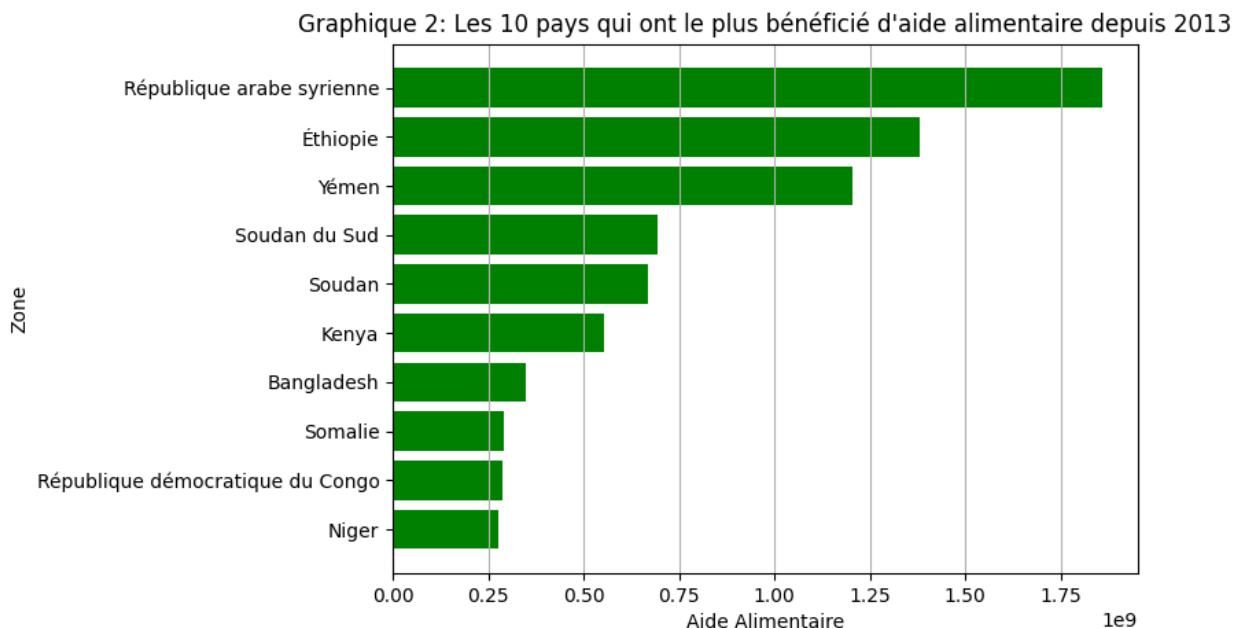
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{"name":"Zone","rawType":"object","type":"string"},{"name":"Aide
Alimentaire","rawType":"int64","type":"integer"}],"conversionMethod":"
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[["0","Afghanistan","185452000"],["1","Algérie","81114000"],
["2","Angola","5014000"],["3","Bangladesh","348188000"],
["4","Bhoutan","2666000"]],"shape":{"columns":2,"rows":5}}

```

```
#affichage après trie des 10 pays qui ont bénéficié le plus de l'aide
alimentaire
total_aide_alimentaire.sort_values(by = ['Aide Alimentaire'],ascending
= False).head(10)

{"columns":[{"name":"index","rawType":"int64","type":"integer"},
{"name":"Zone","rawType":"object","type":"string"},{"name":"Aide
Alimentaire","rawType":"int64","type":"integer"}],"conversionMethod":"
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[["50","République arabe syrienne","1858943000"],
["75","Éthiopie","1381294000"],["70","Yémen","1206484000"],
["61","Soudan du Sud","695248000"],["60","Soudan","669784000"],
["30","Kenya","552836000"],["3","Bangladesh","348188000"],
["59","Somalie","292678000"],["53","République démocratique du
Congo","288502000"],["43","Niger","276344000"]],"shape":
{"columns":2,"rows":10}}

graphique2= total_aide_alimentaire.sort_values(by = ['Aide
Alimentaire'],ascending = False).head(10)
plt.figure(figsize=(7,5))
plt.barh(graphique2['Zone'], graphique2['Aide Alimentaire'], color =
'green')
plt.xticks(fontsize = 10)
plt.yticks(fontsize = 10)
plt.grid(axis='x')
plt.title("Graphique 2: Les 10 pays qui ont le plus bénéficié d'aide
alimentaire depuis 2013")
plt.ylabel('Zone', fontsize = 10)
plt.xlabel('Aide Alimentaire', fontsize = 10)
plt.gca().invert_yaxis()
```



```

#Création d'une liste contenant les 5 pays qui ont le plus bénéficiées
de l'aide alimentaire
cinq_pays = ['République arabe syrienne', 'Éthiopie', 'Yémen', 'Soudan du
Sud', 'Soudan']
cinq_pays

['République arabe syrienne', 'Éthiopie', 'Yémen', 'Soudan du Sud',
'Soudan']

#Création d'un dataframe avec la zone, l'année et l'aide alimentaire
puis groupby sur zone et année
df_cinqpays =
aide_alimentaire.loc[aide_alimentaire['Zone'].isin(cinq_pays),
['Zone', 'Année', 'Aide Alimentaire']]
df_cinqpays.head()

{"columns":[{"name":"index","rawType":"int64","type":"integer"},
{"name":"Zone","rawType":"object","type":"string"},
{"name":"Année","rawType":"int64","type":"integer"}, {"name":"Aide
Alimentaire","rawType":"int64","type":"integer"}], "conversionMethod":"
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[["354", "Éthiopie", "2013", "170000"],
["355", "Éthiopie", "2014", "466000"], ["356", "Éthiopie", "2015", "244000"],
["357", "Éthiopie", "2013", "181066000"],
["358", "Éthiopie", "2014", "178646000"]], "shape":{"columns":3, "rows":5}}

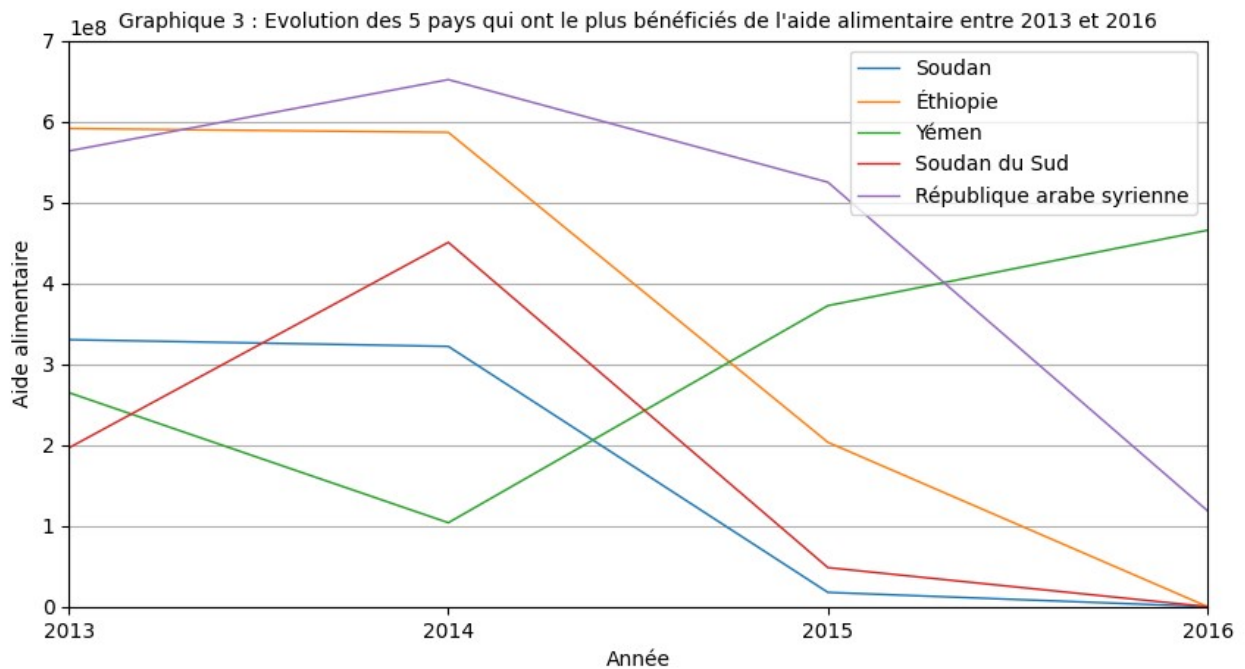
#On filtre sur le dataframe avec notre liste
pivot5 = df_cinqpays.pivot_table(index='Année', columns='Zone',
values='Aide Alimentaire', aggfunc='sum')
pivot5 = pivot5.reset_index()
pivot5.fillna(0, inplace=True)
pivot5

{"columns":[{"name":"index","rawType":"int64","type":"integer"},
{"name":"Année","rawType":"int64","type":"integer"},
{"name":"République arabe
syrienne","rawType":"float64","type":"float"},
{"name":"Soudan","rawType":"float64","type":"float"}, {"name":"Soudan
du Sud","rawType":"float64","type":"float"},
{"name":"Yémen","rawType":"float64","type":"float"},
{"name":"Éthiopie","rawType":"float64","type":"float"}], "conversionMet
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["1", "2014", "651870000.0", "321904000.0", "450610000.0", "103840000.0", "5
86624000.0"],
["2", "2015", "524949000.0", "17650000.0", "48308000.0", "372306000.0", "203
266000.0"],

```

```
[ "3", "2016", "118558000.0", "0.0", "0.0", "465574000.0", "0.0" ]], "shape":
{"columns": 6, "rows": 4}}

# Affichage des pays avec l'aide alimentaire par année
plt.figure(figsize=(10,5))
plt.plot(pivot5['Année'], pivot5['Soudan'], label =
'Soudan',linewidth=1)
plt.plot(pivot5['Année'], pivot5['Éthiopie'], label =
'Éthiopie',linewidth=1)
plt.plot(pivot5['Année'], pivot5['Yémen'], label =
'Yémen',linewidth=1)
plt.plot(pivot5['Année'], pivot5['Soudan du Sud'], label = 'Soudan du
Sud',linewidth=1)
plt.plot(pivot5['Année'], pivot5['République arabe syrienne'], label =
'République arabe syrienne',linewidth=1)
plt.legend(loc='upper right')
plt.xlabel('Année', fontsize=10)
plt.ylabel('Aide alimentaire', fontsize=10)
plt.yticks(fontsize=10)
plt.title("Graphique 3 : Evolution des 5 pays qui ont le plus
bénéficiés de l'aide alimentaire entre 2013 et 2016", fontsize='10')
plt.grid(axis='y')
plt.ylim(0 ,700000000)
plt.xlim(2013,2016)
plt.xticks([2013,2014,2015,2016])
plt.show()
```



```
#Calcul de la disponibilité en kcal par personne par jour par pays
dispo_par_personne_par_pays = dispo_alimentaire.groupby(['Zone'])
['Disponibilité alimentaire (Kcal/personne/jour)'].sum()
dispo_par_personne_par_pays.head()
```

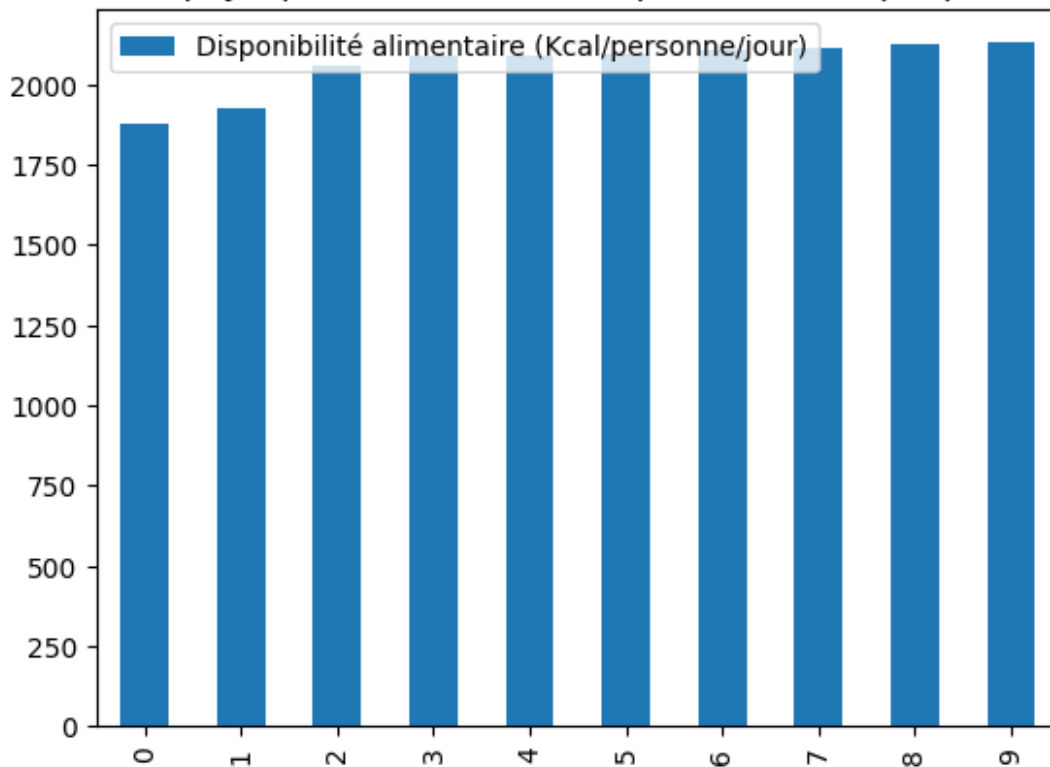
```
Zone
Afghanistan      2087.0
Afrique du Sud    3020.0
Albanie           3188.0
Algérie           3293.0
Allemagne         3503.0
Name: Disponibilité alimentaire (Kcal/personne/jour), dtype: float64
```

```
#Affichage des 10 pays qui ont le moins de dispo alimentaire par
personne
dispo_le_moins = dispo_par_personne_par_pays.sort_values().head(10)
dispo_le_moins = dispo_le_moins.reset_index()
dispo_le_moins
```

```
{"columns":[{"name":"index","rawType":"int64","type":"integer"},
{"name":"Zone","rawType":"object","type":"string"},
{"name":"Disponibilité alimentaire
(Kcal/personne/jour)","rawType":"float64","type":"float"}],"conversion
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1763133c05c6","rows":[["0","République centrafricaine","1879.0"],
["1","Zambie","1924.0"],["2","Madagascar","2056.0"],
["3","Afghanistan","2087.0"],["4","Haïti","2089.0"],["5","République
populaire démocratique de Corée","2093.0"],["6","Tchad","2109.0"],
["7","Zimbabwe","2113.0"],["8","Ouganda","2126.0"],
["9","Éthiopie","2129.0"]],"shape":{"columns":2,"rows":10}}
```

```
#Affichage des 10 pays qui ont le moins de dispo alimentaire par
personne
pays_moins_dispo = dispo_le_moins.sort_values(
    by="Disponibilité alimentaire (Kcal/personne/jour)",
    ascending=True
).head(10)
pays_moins_dispo.plot.bar(title="Les dix pays qui ont le moins de
dispo alimentaire par personne")
plt.show()
```

Les dix pays qui ont le moins de dispo alimentaire par personne



```
#Affichage des 10 pays qui ont le plus de dispo alimentaire par
personne
dispo_le_plus = dispo_par_personne_par_pays.sort_values(ascending =
False).head(10)
dispo_le_plus = dispo_le_plus.reset_index()
dispo_le_plus

{"columns":[{"name":"index","rawType":"int64","type":"integer"},
{"name":"Zone","rawType":"object","type":"string"},
{"name":"Disponibilité alimentaire
(Kcal/personne/jour)","rawType":"float64","type":"float"}],"conversion
Method":"pd.DataFrame","ref":"46ea00e4-d005-4aca-9cc2-
acc4b97e4cbf","rows":[["0","Autriche","3770.0"],
["1","Belgique","3737.0"],["2","Turquie","3708.0"],["3","États-Unis
d'Amérique","3682.0"],["4","Israël","3610.0"],
["5","Irlande","3602.0"],["6","Italie","3578.0"],
["7","Luxembourg","3540.0"],["8","Égypte","3518.0"],
["9","Allemagne","3503.0"]],"shape":{"columns":2,"rows":10}}

#création d'un dataframe avec uniquement la Thaïlande
thaïlande = dispo_alimentaire.loc[dispo_alimentaire['Zone'] ==
'Thaïlande']
thaïlande.head()
```

```

{"columns":[{"name":"index","rawType":"int64","type":"integer"},
{"name":"Zone","rawType":"object","type":"string"},
{"name":"Produit","rawType":"object","type":"string"},
{"name":"Origine","rawType":"object","type":"string"},
{"name":"Aliments pour animaux","rawType":"float64","type":"float"},
{"name":"Autres Utilisations","rawType":"float64","type":"float"},
{"name":"Disponibilité alimentaire
(Kcal/personne/jour)","rawType":"float64","type":"float"},
{"name":"Disponibilité alimentaire en quantité
(kg/personne/an)","rawType":"float64","type":"float"},
{"name":"Disponibilité de matière grasse en quantité
(g/personne/jour)","rawType":"float64","type":"float"},
{"name":"Disponibilité de protéines en quantité
(g/personne/jour)","rawType":"float64","type":"float"},
{"name":"Disponibilité
intérieure","rawType":"float64","type":"float"},{"name":"Exportations
- Quantité","rawType":"float64","type":"float"},{"name":"Importations
- Quantité","rawType":"float64","type":"float"},
{"name":"Nourriture","rawType":"float64","type":"float"},
{"name":"Pertes","rawType":"float64","type":"float"},
{"name":"Production","rawType":"float64","type":"float"},
{"name":"Semences","rawType":"float64","type":"float"},
{"name":"Traitement","rawType":"float64","type":"float"},
{"name":"Variation de
stock","rawType":"float64","type":"float"}],"conversionMethod":"pd.DataFrame",
"ref":"bc510827-5a39-4522-9ae5-06d41bb9a16c","rows":
[["13759","Thaïlande","Abats
Comestible","animale","0.0","0.0","3.0","1.11","0.09","0.56","74000.0",
"5000.0","33000.0","75000.0","0.0","45000.0","0.0","0.0","0.0"],
["13760","Thaïlande","Agrumes,
Autres","vegetale","0.0","0.0","0.0","0.09","0.0","0.0","8000.0","6000
.0","2000.0","6000.0","0.0","12000.0","0.0","2000.0","0.0"],
["13761","Thaïlande","Alcool, non
Comestible","vegetale","0.0","358000.0","0.0","0.0","0.0","0.0","35800
0.0","110000.0","21000.0","0.0","0.0","447000.0","0.0","0.0","0.0"],
["13762","Thaïlande","Aliments pour
enfants","vegetale","0.0","0.0","2.0","0.18","0.01","0.08","12000.0","
7000.0","19000.0","12000.0","0.0","0.0","0.0","0.0","0.0"],
["13763","Thaïlande","Ananas","vegetale","0.0","0.0","10.0","10.02","0
.04","0.08","782000.0","1449000.0","9000.0","671000.0","110000.0","220
9000.0","0.0","0.0","13000.0"]],"shape":{"columns":18,"rows":5}}

```

#Calcul de la sous nutrition en Thaïlande

```

prop_sous_alim_thaïlande = jointure.loc[jointure['Zone'] ==
'Thaïlande']['Prop_sous_alim_par_pays'].sum()
print(f'La proportion de personnes en Thaïlande en sous nutrition est
de {prop_sous_alim_thaïlande:.2f} %')

```

La proportion de personnes en Thaïlande en sous nutrition est de 8.96 %

Disponibilité alimentaire par habitant pour la Thaïlande

```
print('La disponibilité alimentaire par habitant en Thaïlande est de', thaïlande['Disponibilité alimentaire (Kcal/personne/jour)'].sum(), 'Kcal/personne/jour')
```

La disponibilité alimentaire par habitant en Thaïlande est de 2785.0 Kcal/personne/jour

Proportion de la production du manioc sur la production totale de la Thaïlande

```
manioc_total = thaïlande.loc[(thaïlande['Produit'] == 'Manioc')]
print('Le manioc représente', round(manioc_total['Production'].sum()*100/thaïlande['Production'].sum(), 2), '% de toute la production de la Thaïlande')
```

Le manioc représente 14.98 % de toute la production de la Thaïlande

```
print('La Thaïlande produit', round(manioc_total['Production'].sum()), 'Tonnes de Manioc')
```

La Thaïlande produit 30228000 Tonnes de Manioc

Proportion de manioc pour se nourrir

```
print('La Thaïlande utilise seulement', round(manioc_total['Nourriture'].sum()*100/manioc_total['Production'].sum(), 2), '% de sa production de manioc pour se nourrir')
```

La Thaïlande utilise seulement 2.88 % de sa production de manioc pour se nourrir

Proportion de manioc exporté

```
print('La Thaïlande exporte', round(manioc_total['Exportations - Quantité'].sum()*100/manioc_total['Production'].sum(), 2), '% de sa production de manioc')
print('La Thaïlande exporte', round(manioc_total['Exportations - Quantité'].sum()), 'Tonnes de Manioc sur ses', round(manioc_total['Production'].sum()), 'tonnes produits, soit uniquement', (round(manioc_total['Production'].sum()) - round(manioc_total['Exportations - Quantité'].sum())),'tonnes est utilisé pour son pays')
```

La Thaïlande exporte 83.41 % de sa production de manioc

La Thaïlande exporte 25214000 Tonnes de Manioc sur ses 30228000 tonnes produits, soit uniquement 5014000 tonnes est utilisé pour son pays

```
# Répartition de la disponibilité intérieure du manioc

repartition_disponibilite_intérieure_manioc = []

Liste_utilisation_produits_thaïlande = ['Aliments pour animaux',
'Pertres', 'Nourriture', 'Autres Utilisations']

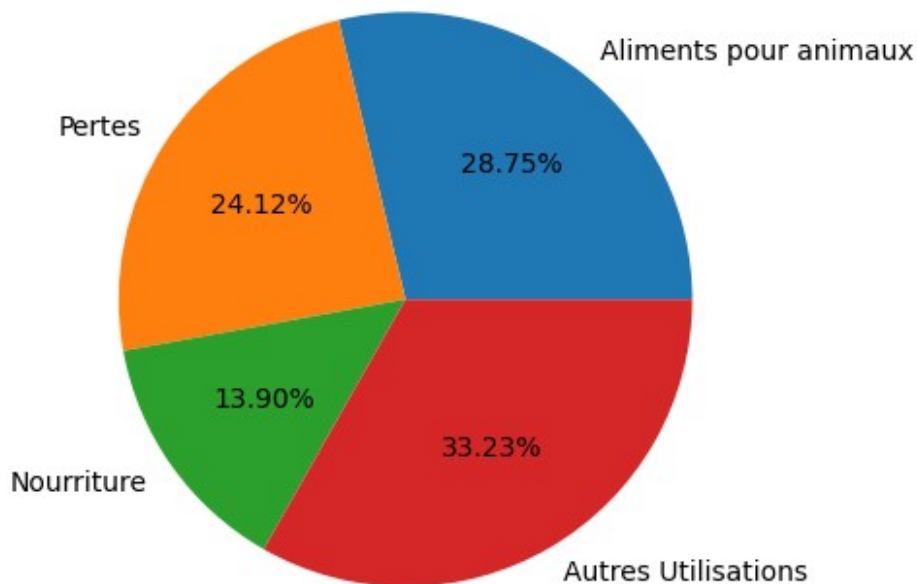
for i in Liste_utilisation_produits_thaïlande:
    proportion = round(manioc_total[i].sum() * 100 /
manioc_total['Disponibilité intérieure'].sum(), 2)
    print('Proportion de manioc pour', i, ':', proportion, '%')
    repartition_disponibilite_intérieure_manioc.append(proportion)
```

```
Proportion de manioc pour Aliments pour animaux : 28.74 %
Proportion de manioc pour Pertres : 24.12 %
Proportion de manioc pour Nourriture : 13.9 %
Proportion de manioc pour Autres Utilisations : 33.22 %
```

Graphique 4:

```
plt.pie(repartition_disponibilite_intérieure_manioc, labels =
Liste_utilisation_produits_thaïlande, autopct='%1.2f%%')

([<matplotlib.patches.Wedge at 0x1b4378446e0>,
<matplotlib.patches.Wedge at 0x1b4378461b0>,
<matplotlib.patches.Wedge at 0x1b4355074d0>,
<matplotlib.patches.Wedge at 0x1b435507470>],
[Text(0.6811187685342922, 0.8637576182878676, 'Aliments pour
animaux'),
Text(-0.9215888958129591, 0.6005613266888328, 'Pertres'),
Text(-0.8970954929517103, -0.6365686738488849, 'Nourriture'),
Text(0.553190181957659, -0.9507789556914121, 'Autres
Utilisations')],
[Text(0.3715193282914321, 0.47114051906610954, '28.75%'),
Text(-0.502684852261614, 0.327578905466636, '24.12%'),
Text(-0.48932481433729647, -0.34721927664484625, '13.90%'),
Text(0.3017400992496322, -0.5186067031044066, '33.23%')])
```



1 - La Thaïlande est un important producteur de manioc, et la demande mondiale pour ce produit peut inciter le pays à exporter une grande partie de sa production

2 - Le manioc est aussi utilisé pour d'autres applications, comme l'alimentation animale, produits industriels et autres

3 - Les politiques gouvernementales peuvent influencer les décisions d'exportation et de consommation intérieure

#Rajouter en dessous toutes les analyses complémentaires suite à la demande de mélanie :

#"et toutes les infos que tu trouverais utiles pour mettre en relief les pays qui semblent être

#le plus en difficulté au niveau alimentaire"